

Advanced (Habanero)

Q1. What is the decimal equivalent of $\frac{3}{4}$?

- (A) 0.25
- (B) 0.5
- (C) 0.75
- (D) 0.8
- (E) 0.9

Q2. Which of the following is equal to 0.6?

- (A) $\frac{3}{5}$
- (B) $\frac{6}{10}$
- (C) $\frac{3}{10}$
- (D) $\frac{2}{3}$
- (E) $\frac{1}{2}$

Q3. Compare the rational numbers $-\frac{2}{3}$ and $-\frac{1}{2}$. Which one is greater?

- (A) $-\frac{2}{3}$
- (B) $-\frac{1}{2}$
- (C) equal
- (D) less than
- (E) greater than

Q4. What is $2 \cdot \frac{3}{4}$?

- (A) $\frac{1}{2}$
- (B) $\frac{3}{2}$
- (C) $\frac{5}{4}$
- (D) $\frac{6}{4}$
- (E) $\frac{8}{4}$

Q5. What is the value of x in $\frac{1}{2}x = \frac{3}{4}$?

- (A) $\frac{3}{2}$
- (B) $\frac{2}{3}$
- (C) $\frac{3}{4}$
- (D) $\frac{4}{3}$
- (E) $\frac{6}{4}$

Q6. What percent of 200 is 50?

- (A) 10
- (B) 20
- (C) 25
- (D) 30
- (E) 40

Q7. What is $\frac{1}{8}$ as a percent?

- (A) 5
- (B) 10
- (C) 12.5
- (D) 20
- (E) 25

Q8. Which of the following rational numbers has the greatest value on the number line?

- (A) $-\frac{1}{2}$
- (B) $\frac{1}{4}$
- (C) $\frac{1}{3}$
- (D) $\frac{2}{3}$
- (E) $\frac{3}{4}$

Open Ended Questions (FRQ)

Q9. Show that $\frac{5}{6} + \frac{1}{8} = \frac{23}{24}$ and explain your steps.

Q10. Given the rational number $-\frac{3}{4}$, represent it on a number line, and then find its decimal equivalent. Explain the process.

Chef's Tips

- Please ensure students have scratch paper and a calculator.
- Remind students to show their work and explain their reasoning.
- Emphasize the importance of using the correct operations when solving problems.